# Category X Ranking Algorithm v14

Daily scoring, persistence-aware adaptive movement, and worked examples

## Purpose

The engine ranks games every day. Daily signals describe how strongly a game is performing. The final score produces an ideal rank. A separate movement layer decides how quickly the website position should move toward that ideal rank. This prevents short-term noise from causing large position swings while still allowing persistent improvements to surface.

## Current scoring weights

| Component | Weight | Role |
|---|---|---|
| Current Performance | 65% | Measures recent game strength |
| Momentum | 15% | Measures direction and strength of the trend |
| Acceleration | 5% | Measures whether the momentum is strengthening or weakening |
| Stability | 15% | Measures consistency of the recent pattern |

## 1. Current Performance

Current Performance uses the most recent 7 days for each game. The newest observations have greater influence than the oldest. Four performance measures are combined: Clients, Bet Count, Stakes and GGR. Each measure is normalized relative to the catalogue so the four measures can be combined on a 0 to 100 scale.

### Worked example

Suppose a game receives normalized estimates of Clients = 96, Bet Count = 99, Stakes = 99 and GGR = 98. Using the current metric weights of 20%, 15%, 30% and 35% gives: 96 x 0.20 + 99 x 0.15 + 99 x 0.30 + 98 x 0.35 = 98.05. Current Performance = 98.05/100.

## 2. Momentum

Momentum compares recent activity with preceding activity over 3-day, 7-day and 14-day windows. The 3-day component is more responsive, the 7-day component provides the main recent trend, and the 14-day component provides longer context. The resulting value is signed: positive means improving, negative means declining.

Example: a game can have 3-day momentum of +40, 7-day momentum of +25 and 14-day momentum of +10. The combined momentum is positive because all three windows point upward, but the short-term result is stronger than the longer-term result.

## 3. Acceleration

Acceleration measures how the momentum is changing. Positive acceleration means the trend is becoming more positive. Negative acceleration means the trend is becoming less positive. A game may therefore have negative momentum and positive acceleration: it is still declining, but the decline is slowing or beginning to recover.

Example: momentum changes from -30 to -10. Acceleration is positive because momentum moved 20 points in a positive direction. The game is still below zero, so it is not yet an improving game.

## 4. Stability

Stability measures how consistent recent performance has been. A stable game follows a more repeatable pattern. A volatile game has large swings. Stability is used as a confidence signal and contributes 15% to the final score in v14.

## 5. Final ranking score

The final score is the weighted combination of Current Performance, Momentum, Acceleration and Stability. Acceleration is intentionally kept at 5% after sensitivity testing showed that larger weights reduced Top-10 and Top-20 overlap on the current evaluation data.

### Example

If Current Performance = 90, Momentum = 70, Acceleration = 60 and Stability = 80, the final score is 90 x 0.65 + 70 x 0.15 + 60 x 0.05 + 80 x 0.15 = 83.5. The games are then sorted by this score to obtain the ideal rank.

## 6. Ideal rank versus website position

Ideal rank is the position implied by the final score. Website position is the actual simulated position shown to users. The two can differ because the website position moves toward the ideal rank under controlled movement rules.

## 7. Persistence-aware adaptive movement

Persistence is a movement multiplier, not a fixed waiting period and not a separate ranking-score component. The engine first measures movement pressure from evidence strength, momentum and acceleration. Persistence then increases the amount of that pressure that is allowed to affect the website position. Strong established games also receive protection from abrupt downward replacement.

The v14 upward movement concept can be summarized as: base movement pressure x persistence multiplier. The multiplier increases with the direction streak, but only when the underlying signal strength is meaningful. This concentrates larger movements on games that have both strong evidence and sustained direction.

## 8. Worked movement example

A game starts at #200 and has an ideal rank of #60. On Day 1 it has strong evidence but only one day of persistence, so the allowed move is modest. If momentum and acceleration remain strong for Days 2, 3 and 4, the persistence multiplier grows and the allowed movement increases. The game can therefore move progressively from #200 toward #60 instead of jumping directly to #60 or moving only one position per day.

The important rule is: a short-lived spike can move a game, but a sustained and strengthening signal can move it faster.

## 9. Missing data

Missing data is not treated as zero performance. A game with no selected-day record is labelled No Data, its raw metrics show No Data, and its website position is held rather than being moved because of an artificial zero.

## 10. V14 real-data validation

| Measure | V13 baseline | V14 targeted multiplier |
|---|---|---|
| Spearman correlation | 0.6791 | 0.6816 |
| Top-10 overlap | 52.3% | 51.7% |
| Top-20 overlap | 55.0% | 54.0% |
| Top-100 overlap | 76.3% | 76.2% |
| Mean rank error | 380.1 | 377.2 |
| Average movement | 6.08 | 7.02 |

| Measure | V13 baseline | V14 targeted multiplier |
|---|---|---|
| Median movement | 4 | 5 |
| Maximum movement | 32 | 36 |
| Moves >20 positions | 2.33% | 5.72% |
| Direction reversals | 24.89% | 24.72% |

Interpretation: the targeted multiplier increases responsiveness while keeping overall ranking quality close to the baseline. High-evidence games with at least four days of persistence moved about 12.8 positions on average in the real-data test. The trade-off is a higher rate of moves above 20 positions, so continued monitoring is still required.